CHAPTER 1
REPRODUCTION IN ORGANISMS

Q.1 Mention the sites where syngamy occurs in amphibians and reptiles respectively. **(C.B.S.E 2010)**

Q.2 Offspring derived by asexual reproduction are called clones. Justify giving two reasons. **(C.B.S.E 2010)**

Q.3 Mention the characteristic features and a function of zoospores in some algae. **(C.B.S.E 2010)**

Q.4 Name an organism where cell division is itself a mode of reproduction. **(C.B.S.E 2010)**

Q.5 In Yeast and Amoeba, the parent cell divides to give rise to two new individual cells. How does the cell division differ in these two organisms? **(C.B.S.E 2010)**

Q.6 Name the type of cell division that takes place in the zygote of an orgnaism exhibiting haplontic life cycle. **(C.B.S.E 2010)**

Q.7 How does Penicillium reproduce asexually? **(C.B.S.E 2011)**

Q.8 Name the phase all organisms have to pass through before they can reproduce sexually. **(C.B.S.E 2011)**

Q.9 Name the group of orgnisms that produces non-motile male gametes. How do they reach the female gamete for fertilization **(C.B.S.E 2011)**

Q.10 Mention the unique flowering phenomenon exhibited by Strobilantheskunthiana. **(C.B.S.E 2012)**

Q.11 Cucurbits and Papaya plants bear staminate and pistillate flowers. Mention the categories they are put separately on the basis of type of flowers they bear. **(C.B.S.E 2012)**

Q.12 Give reason : some organisms like honeybees are called parthengenetic animals. **(C.B.S.E 2012)**

Q.13 A moss plant produces a large number of antherozoids but relatively only a few egg cells. Why? **(C.B.S.E 2010)**

Q.14 Why are Papaya and Data Palm plants said to be dioecious whereas cucurbits and coconut palms are monoecious, inspite of all of them bearing unisexual flowers ? **(C.B.S.E 2010)**

Q.15 The cell division involved in gamete formation is not of the same type in different orgnaims justify **(C.B.S.E. 2011)**

Q.16 A list of three flowering plants is given below. Which ones out of them are
(i) monoecious and
(ii) bearingpistillate flowers. Date Palm, Cucurbits, Pea. **(C.B.S.E 2011)**

Q.17 (a) State the difference between meiocyte and gamete with respect to chromosome number.
(b)Why is whiptail lizard referred to as parthenogenetic? **(C.B.S.E, 2012)**

18.(a) Coconut palm is monoecious, while date palm is dioecious. Why are they so called?
(b) Draw a labelled diagram of sectional view of a mature embryo sac of an angiosperm.**(2014)**
19. Why do algae and fungi shift to sexual mode of reproduction just before the onset of adverse conditions? **.(2014)**

CHAPTER 2
SEXUAL REPRODUCTION IN FLOWERING PLANTS

Q.1 Write briefly the role of pollination in the growth and development in an angiosperm. **(C.B.S.E 2007)**

Q.2 Describe the structure of a typical/polygonum type embryo sac found in flowering plants. Why is it called monosporic? **(C.B.S.E 2007)**

Q.3 Why is the process of fertilization in a flowering plant referred to as double fertilization? **(C.B.S.E 2007)**

Q.4 What is the process of fertilization in flowering plant referred to as double fertilization ? **(C.B.S.E 2007)**

Q.5 The flower of Brinajal is referred to as chasmoganous while that of Bean is cleistogamous. How are they different from each other.

Q.6 Coconut Palm is monoecious while Date Palm is dioecious. Why are they called so ? **(C.B.S.E 2008)**

Q.7 Banana is a parthenocarpic fruit whereas oranges show polyembryony. How are they different from each other with respect to seeds? **(C.B.S.E 2009)**

Q.8 Name the cell from which the endosperm of Coconut develops. Give the characteristic features of endosperm of coconut. **(C.B.S.E 2009)**

Q.9 Draw a vertical section of a Maize grain and label.
(i) pericarp (ii) scutellum (iii) coleoptile (iv) radicle **(C.B.S.E 2009)**

Q.10 Fertilization is essential for production of seeds

(i) Give one example of an angiosperm that produces seed without fertilization. Name the process.

(ii) Explain two ways by which seeds develop without fertilization. **(C.B.S.E 2009)**

Q.11 Explain any two devices by which autogamy is prevented in flowering plants. **(C.B.S.E 2009)**

Q.12 Mention the reasons for difference in ploidy of zygote and primary endosperm nucleus in an angiosperm . **(C.B.S.E 2010)**

Q.13 How does the floral pattern of Mediterranean orchid, Ophrys, guarantee cross pollination ? **(C.B.S.E 2010)**

Q.14 Draw a longitudnal section of a post pollinated pistil to who entry of pollen tube into mature embryo sac. Label filiform apparatus, chalazal end, hilum, antipodals, male gametes and secondary nucleus. **(C.B.S.E 2010)**

Q.15 Where does triple fusion take place in a flowering plant. Why is it so called ? Mention its significance. **(C.B.S.E 2010)**

Q.16 If you squeeze a seed of orange, you might observe many embryos of different sizes. How is it possible ? Explain. **(C.B.S.E 2010)**

Q.17 (a) Mention any four strategies adopted by flowering plants to prevent self pollination.

(b) Why is geitonogamy also referred to geneticalautogamy ? **(C.B.S.E 2010)**

Q.18 Explain giving two reasons why pollen grains can be best measured as fossils. **(C.B.S.E 2010)**

Q.19 How many haploid cells are present in a mature female gametophyte of a flowering plant. Name them. **(C.B.S.E 2010)**

Q.20 Differentiate between albuminous and non-albuminous seeds, giving one example each. **(C.B.S.E 2010)**

Q.21 Draw a diagram of a male gametophyte of an angiosperm. Label any four parts . Why is sporollenin considered the most resistant organic material ? **(C.B.S.E 2011)**

Q.22 Differentiate between geitonogamy and xenogamy in plants. Which one between the two will lead to inbreeding depression and why ? **(C.B.S.E 2011)**

Q.23 Where is sporollenin present inplants ?state its significance with reference to its chemical nature. **(C.B.S.E. 2012)**

Q.24 State one advantage and one disadvantage of cleistogamy. **(C.B.S.E. 2012)**

Q.25 Explain the function each of 9a) Coleorhiza (b) Germ pores . **(C.B.S.E 2012)**

Q.26 How does the study of different parts of a flower help in identifying wind as its pollinating agent ? **(C.B.S.E 2012)**

Q.27 Write the cellular contents carried by the pollen tube. How does the pollen tube gain entry into the embryo sac ? **(C.B.S.E 2012)**

Q.28 Name the product of fertilization that forms the kernel of coconut. How does the Kernel differ from coconut water ? **(C.B.S.E. 2012)**

Q.29 (a) Mention the similarity between autogamy and geitonogamy.
(b) How does geitonogamy differ from xenogamy ? **(C.B.S.E 2012)**

Q.30 Differentiate perisperm and endosperm giving one example of each . **(C.B.S.E 2012)**

Q.31 How does pollen mother cell develop into a mature pollen grain ? Illustrate the stages with labelled diagrams. **(C.B.S.E 2009, 2010)**

Q.32 (a) Draw a labelled diagram of a mature embryo sac. **(C.B.S.E 2009, 2010)**
(b) Why does a pollen grain possess two male gametes ? **(C.B.S.E 2009)**

Q.33 (a) Trace the development of embryo after synagmy in a dicot plant.
(b) Endosperm development precedes embryo development. Explain.
(c) Draw a diagram of a mature dicot embryo and label cotyledons, plumule, radicle, hypocotyl in it. **(C.B.S.E 2009)**

Q.34 Describe in sequence the events that lead to the development of a 3-celled pollen grain from microspore mother cell in angiosperms. **(C.B.S.E.2010)**

Q.35 (a) Draw a labelledlongisectional view of an alubumniousseed ? **(C.B.S.E 2010)**
(b) How are seeds advantageous to flowering plants ?

Q.36 Explain double fertilization and trace the post fertilization events in sequential order leading to seed formation in a typical dicotyledonous plant. **(C.B.S.E 2010)**

Q.37 Give reasons why

(i) Most zygotes in angiosperms divide only after certain amount of endosperm is formed.

(ii) Groundnut seeds are exalbuminous and Caster seeds are albuminous.

(iii) Micropyle remains as a small pore in the seed coat of a seed.

(iv) Integuments of an ovule harden and the water content is highly reduced as the seed matures.

(v) Apple and Cashew are not called true fruits. **(C.B.S.E. 2011)**

Q.38 (a) Draw a labelled diagram of L.S. on an embryo of grass (any six labels).
(b) Give reasons for each of the following **(C.B.S.E. 2011)**

Q.39 (a) Draw a diagram of an enlarged view of T.S. one microsporangium of an angiosperm and label the following parts :
(i) Tapetum (ii) Middle layer
(iii) Endothecium (iv) Microspore mother cells

(b) Mention the characteristic features and functions of tapetum.

(c) Explain the following giving reasons :

(i) Pollen grains are well preserved as fossils

(ii) Pollen tablets are in use by people these days. **(C.B.S.E. 2011)**

Q.40 (a) Why is the process of fertilization in angiosperms termed as double fertilization. Explain.

(b) Draw a diagram of an angiospermic embryo sac where fertilization is just completed. Label the following (i) Micropylar and of embryo sac (ii) Part that develops into an embryo (iii) Part that develops into an endosperm (iv) The degenerating cells at chalazal end.

(c) Draw a labelled diagram of globular embryonic stage of an angiosperm. **(C.B.S.E. 2011)**

Q.41 (a) Explain the characteristic features of wind pollinated flowers. How are insect pollinated followers different from them ?

(b) Explain the mutually rewarding relationship between Yucca plant and species of moth. **(C.B.S.E. 2011)**

Q.41 How does the megaspore mother cell develop into 7-celled, 8 –nucleate embryo sac in an angiosperm ?

Draw labelled diagram of a mature embryo sac. **(C.B.S.E. 2012)**

Q.42 (a) Why is fertilization in an angiosperm referred to as double fertilization ? Mention the policy of the cells involved.

(b) Draw a neat labelled sketch of L.S. of an endospermous monocot seed. **(C.B.S.E. 2012)**

Q.43 (a) How does microspore mother cell develop into mature pollen grain in angiosperms ?

(b) Describe the structure of a mature pollen grain and draw a labelled diagram of its two celled stage. **(C.B.S.E. 2012)**

44.Why are beehives kept in a crop field during flowering period? Name any two crop fields where this is practised. **(C.B.S.E. 2014)**

45.Explain any three advantages that seeds offer to angiosperms.
(C.B.S.E. 2014)

45.(a) Why does endosperm development precede embryo development in angiosperm seeds? State the role of endosperm in mature albuminous seeds.

(b) Describe with the help of three labelled diagrams the different embryonic stages that include mature embryo of dicot plants **(C.B.S.E. 2014)**

CHAPTER-3
HUMAN REPRODUCTION

1. Why does failure of testes to descend into scrotum produce sterility? (C.B.S.E. 2006 Comptt.)
2. Sperms have a tail whereas eggs do not. Why so? (C.B.S.E. 2007)
3. Mention the function of trophoblast in human embryo. (C.B.S.E. 2011)
4. Name of embryonic stage that gets implanted in the uterine wall of a human female. (C.B.S.E. 2011)
5. What stimulates pituitary to release the hormone responsible for parturition? Name the hormone. (C.B.S.E. 2011)
6. List the changes the primary oocyte undergoes in the tertiary follicular stage in the human ovary. (C.B.S.E. 2012)
7. Write the location and function of Sertoli cells in humans. (C.B.S.E. 2012)
8. When do the oogenesis and spermatogenesis initiate in human females and males respectively? (C.B.S.E. 2012)
9. Mention the difference between spermiogenesis and spermiation. (C.B.S.E. 2012)
10. Where is acrosome present in humans? Write its functions. (C.B.S.E. 2012)
11. Explain the function of umbilical cord. (C.B.S.E. 2012)
12. How is the entry of only one sperm and not many ensured into an ovum during fertilization in humans? (C.B.S.E. 2012)
13. Identify the given figure and the part labeled 'A' (C.B.S.E. 2012)
14. Mention the location and function of Leydig cells in humans. (C.B.S.E. 2012)
15. Mention the function of mitochondria in sperm. (C.B.S.E. 2012)
16. What is monospermy? How is polyspermy prevented in humans. (C.B.S.E. 2007)
17. What is gynaecomastia ? What is its cause during neonatal period and during puberty? (C.B.S.E. 2007
18. What is pregnancy hormone? Why is it so called? Name two sources of the hormone in a human female. (C.B.S.E. 2007
19. Draw a labeled diagram of ovum and label its four parts. (C.B.S.E. 2007
20. Why is the human placenta referred to as haemochorial type? Explain. (C.B.S.E. 2008)
21. Draw a labeled diagram of a sectional view of human ovary showing various stages of follicles growing in it. (C.B.S.E. 2008)
22. Where are fimbriae present in a human female reproductive system? Give their function. (C.B.S.E. 2009)
23. Name the muscular and glandular layers of human uterus. Which one of these layers undergoes cyclic changes during menstrual cycle? Name the hormone essential for maintenance of this layer.(C.B.S.E. 2009)
24. Where are Leydig cells present ? What is their role in reproduction?

(C.B.S.E. 2009)

25. Name the source of gonadotropins in human females. Explain the changes brought about in the ovary by these hormones during menstrual cycle. (C.B.S.E. 2009)
26. Placenta acts as an endorine tissue. Justify. (C.B.S.E. 2010)
27. Mention the fate of corpus luteum and its effects on the uterus in the absence of fertilization of the ovum in the human female. (C.B.S.E. 2010)
28. Spermatogenesis in human males is a hormone regulated process. Justify. (C.B.S.E. 2010)
29. Draw a labeled diagram of the reproductive system in human female. (C.B.S.E. 2011)
30. Name the hormones produced only during pregnancy in human female. Mention their source organ. (C.B.S.E. 2011)
31. (a) Where do the signals for parturition originate in humans?
    (b) Why is it important to feed the new born babies on colostrum (C.B.S.E. 2012)
32. (a) Oviducal fimbriae (b) Oxytocin
    (c) Middle piece in human sperm (d) Seminal vesicle
    (e) Acrosome of human sperm (C.B.S.E. 2012)
33. Give a schematic representation of of oogenesis in humans. Mention the number of chromosomes at each stage. Correlate the life phases of the individual with the stages of the process (C.B.S.E. 2008)
34. (a) Give a schematic representation of spermatogenesis in humans.
    (b) At which stage of life does gametogenesis begin in human male and femalerespectively?
    (c) Name the organs where gametogenesis gets completed in human male and female respectively. (C.B.S.E. 2008)
35. (a) Draw a labeled diagram of a sectional view of human seminiferous tubule.
    (b) Differentiate between gametogenesis in human males and females on the basis of (i) Time of initiation of the process. Products formed at the end of the process.(C.B.S.E. 2008)
36. Draw a diagrammatic sectional view of human ovary showing different stages of oogenesis alongwith corpus luteum. (C.B.S.E. 2009)
37. Where is morula formed in humans? Explain the process of its development from zygote.(C.B.S.E. 2009)
38. (a) Draw a schematic diagram of human sperm and label the cellular componentsgive the functions of any three parts.
    (b) Where are the sperm heads found embedded after spermatogenesis? (C.B.S.E. 2009)
39. (a) When and how does placenta develop in human female?
    (b) How is placenta connected to the embryo?
    (c) Placenta acts as an endocrine gland. Explain. (C.B.S.E. 2009)
40. When and where are primary oocytes formed in human female? Trace the development of these oocytes till ovulation (in menstrual cycle). How do gonadotropins influence this developmental process? (C.B.S.E. 2010)
41. (a) Explain the events taking place at the time of fertilization of an ovum in a human female.
    (b) Trace the development of zygote upto its implantation in the uterus.
    (c) Name and draw a labeled sectional view of the embryonic stage that gets implanted. (C.B.S.E. 2010)
42. (a) Give a schematic representation showing the events of spermatogenesis in human male.
    (b) Describe the structure of a human sperm. (C.B.S.E. 2010)
43. (a) Draw a diagrammatic labeled sectional view of a seminiferous tubule of a human

(b) Describe in sequence the process of spermatogenesis in humans.

(C.B.S.E. 2010)

44. Describe the post-zygotic events leading to implantation and placenta formation in humans. Mention any two functions of placenta. (C.B.S.E. 2010)

45. Explain the process of fertilization in human female. Trace the post-fertilization events in sequential order upto implantation of the embryo.

(C.B.S.E.2010)

46. (a) Draw a labeled diagram of the human female reproductive system.
    (b) Enumerate the events in the ovary of a human female during (i) Follicular phase (ii) Lufeal phase of menstrual cycle. (C.B.S.E. 2010)

47. (a) Write the specific location and functions of the following cells in human males.
    (i) Leydig cells (ii)Sertoli cells (iii) Primary spermatocyle
    (b) Explain the role of two accessory glands in human male reproductive system.

(C.B.S.E. 2011)

48. (a) When and where does spermatogenesis occur in a human male?
    (b) Draw a diagram of a mature human male gamete. Label the following parts : acrosome, nucleus, middle piece, tail.
    (c) Mention the function of acrosome and middle piece.

(C.B.S.E. 2011)

49. The following is the illustration of sequence of ovarian events (a-i) in human female?

DIAGRAM

(i) Identify the figure that illustrates ovulation and mention the stage of oogenesis it represents.
(ii) Name the ovarian hormone and pituitary hormone that has caused the above mentioned event.
(iii) Explain the changes that occur in the uterus simultaneously in anticipation.
(iv) Write the difference between 'c' and 'h'.
(v) Draw a labeled sketch of human ovum prior to fertilization.

OR

(i) Identify the figure that illustrates corpus luteum and name the pituitary hormone that influences its formation.
(ii) Specify the endocrine function of corpus luteum. How does it influence uterus? Why is it essential?
(iii) What is difference between 'd' and 'e' ?
(iv) Draw a labeled sketch of Graafian follicle. (C.B.S.E. 2012)

50. Study the illustration and answer the questions

DIAGRAM

    (i) Identify 'a'
    (ii) Name and state the function of 'c'
    (iii) Identify 'd'
    (iv) Explain the role of hormones in the formation and release of 'a'
    (v) Draw a diagram of 'b' separately and label the parts : that helps its entry into 'a'; that carries genetic material ; that helps in its movements.

(C.B.S.E.2012)

51. (a) How is 'oogenesis' markedly different from 'spermatogenesis' with respect to the growth till puberty in the humans?
    (b) Draw a sectional view of human ovary and label the different follicular stages, ovum and Corpus luteum. **2014**

52.Name and explain the role of the inner and middle walls of the human uterus. **2014**

53.Explain the ovarian and uterine events that occur during a menstrual cycle in a human female under the influence of pituitary and ovarian hormones respectively. **2014**

C.B.S.E. BOARD QUESTIONS

1. Name and explain the role of inner and middle wall of human uterus?
   Women are often blamed for producing female children. Consequently, they are ill-treated and ostracized. How will you address the issue scientifically?
2. A) How is oogenesis markedly different from spermatogenesis with respect to growth till puberty?
3. B) Draw sectional view of human ovary. Label follicular stages, ovum, and corpus luteum.
4. Draw labeled diagram of sectional view of seminiferous tubule (label any six parts)
5. Explain ovarian and uterine events during menstrual cycle under the influence of pituitary and ovarian hormones.
6. A) differentiate between: i)vas deferens and vas efferentia
   ii) Spermatogenesis and spermiogenesis
   b) Draw labeled diagram of male reproductive system.

CHAPTER-4
REPRODUCTIVE HEALTH

1. Mention any two events that are inhibited by the intake of oral contraceptive pills to prevent pregnancy in humans. (C.B.S.E. 2009 Comptt.
2. Why is tubectomy considered a contraceptive method? (C.B.S.E. 2010)
3. A mother of one year old daughter wanted to space her second child. Her doctor suggested Cu-T. Explain its contraceptive actions. (C.B.S.E. 2008)
4. (a) Expand IUD (b) Why is hormone releasing IUD considered a good contraceptive to space children? (C.B.S.E. 2008)
5. Explain any two methods of Assisted Reproductive Technology (ART) that has helped childless couples to bear children. (C.B.S.E. 2008 Comptt.)
6. How does Cu T act as an effective contraceptive for human females? (C.B.S.E. 2009)
7. Name the hormonal composition of oral contraception used by human females. Explain how does it act as contraceptive? (C.B.S.E. 2009)
8. Why do some women use " Saheli" pills? (C.B.S.E. 2009)
9. How are assisted reproductive technologies helpful to humans? How are ZIFT and GIFT different from intrauterine transfers ? Explain. (C.B.S.E. 2009)
10. Name any two copper releasing intra-uterine devices (IUDs). List two reasons that make them effective contraceptives. (C.B.S.E. 2009 Comptt.)
11. How do copper and hormone releasing IUDs act as contraceptives ?Exlain. (C.B.S.E. 2010)
12. Explain the zygote intrafallopian transfer technique (ZIFT). How is intrauterine transfer technique (IUT) different from it? (C.B.S.E. 2010)
13. What is amniocentesis ? Why has the government imposed a statutory ban inspite of its importance in the medical field? (C.B.S.E. 2010)
14. Why is 'Saheli' a well accepted contraceptive pill? (C.B.S.E. 2010)
15. Why is CuT considered a good contraceptive device to space children? (C.B.S.E. 2011)
16. Name an oral pill used as a contraceptive by human females. Explain how does it prevent pregnancy? (C.B.S.E. 2011)
17. At the time of independence, the population of India was 350 million which exploded to over 1 billion by May 2000. List any two reasons for this rise in population and any two steps taken by the government to check this population explosion. (C.B.S.E. 2011)
18. Explain how do the following act as contraciptives (a)CuT(b)'Saheli' (C.B.S.E. 2012)

19.Women are often blamed for producing female children. Consequently, they are ill-treated and ostracised. How will you address this issue scientifically if you were to conduct an awareness programme to highlight the values involved?_ (C.B.S.E. 2014)